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JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

An Autonomous Institute, with NAAC "A" Grade

Basic Science & Humanities Department

2023-24 (EVEN Semester)

Education to Eternity

MID SEM EXAMINATION

Course: B. Tech.

Subject: Basics of Civil & Mechanical Engineering

Subject Code: CSE/DS/AI/2T004

Year/Semester: IInd Semester (First Year)

Max Marks: 20 **Date :- 23/04/2024** **Duration:- 1 Hr**

Instructions to the Students:

1. All the questions carry marks as indicate.

2. Assume suitable data whenever necessary.

3. Illustrate your answer whenever necessary with the help of neat sketches.

CO1 Define basic stream of Mechanical & Civil Engineering.

CO2 Explain the concepts of product manufacturing, Energy engineering, design engineering, Automobile engineering, construction technique and civil surveying.

CO3 Apply Basic knowledge of Casting, Machining, Designing, Manufacturing & Civil Construction technique.

CO4 Analysed the different mechanical system and properties of construction & surveying material.

CO5 Interpret the problem in mechanical system and civil structure.

SECTION A

1. All Questions are compulsory.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define factor of safety?	1/CO1	1
Q.2	Define machinability.	2/CO1	1
Q.3	What is the function of tailstock?	1/CO2	1
Q.4	What the function of flywheel in reciprocating IC engine.	2/CO2	1

SECTION B

1. All Questions are compulsory.

Q.5	What are the Non-conventional energy source?	1/CO1	2
Q.6	Enlist the main parts of carriage assembly of lathe machine.	2/CO2	2
Q.7	Write basic elements of machining with neat sketch.	1/CO1	2
Q.8	How vehicles are classified on the basis of Fuel used?	2/CO2	2

SECTION C

1. Solve any two questions.

Q.9	Compare between two-stroke and four-stroke engine?	3/CO3	4
Q.10	Categories the Drilling, Reaming and Boring operations?	3/CO3	4
Q.11	State the general consideration in Machine Design.	4/CO4	4